

KAZAN, Russian Federation

WMO: 275950

Lat: 55.7400N

Lon: 49.2044E

Elev: 119

StdP: 99.91

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.6	-24.2	-30.9	0.2	-27.5	-27.3	0.3	-23.9	12.1	-7.0	11.0	-5.5	3.4	140	0.644

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	31.1	19.8	29.1	19.3	27.2	18.5	21.3	27.8	20.4	26.7	19.5	25.5	3.7	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.1	14.1	24.3	18.1	13.2	23.1	17.2	12.5	22.4	62.6	27.8	59.3	26.7	56.3	25.4	26.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.6	9.6	8.8	-29.8	33.2	3.3	2.0	-32.2	34.7	-34.2	35.8	-36.0	37.0	-38.4	38.4		
(5)			-30.0	23.1	3.3	1.3	-32.4	24.0	-34.3	24.8	-36.2	25.6	-38.5	26.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	4.7	-10.8	-10.6	-4.3	5.4	13.5	17.8	20.2	18.2	12.3	5.0	-2.7	-8.4	(6)
(7)		DBStd	12.42	7.07	6.96	4.75	5.38	5.08	4.30	3.69	4.24	4.39	4.35	5.66	6.90	(7)
(8)		HDD10.0	2991	646	578	443	158	24	2	0	1	25	164	380	570	(8)
(9)		HDD18.3	5169	904	811	702	387	165	61	23	54	188	414	630	828	(9)
(10)		CDD10.0	1059	0	0	0	21	131	234	316	256	93	8	0	0	(10)
(11)		CDD18.3	196	0	0	0	0	14	44	81	51	6	0	0	0	(11)
(12)		CDH23.3	1753	0	0	0	2	150	391	709	462	38	0	0	0	(12)
(13)		CDH26.7	538	0	0	0	0	29	116	226	162	5	0	0	0	(13)
(14)	Wind	WSAvg	4.1	4.5	4.3	4.3	4.1	4.3	3.8	3.4	3.7	3.8	4.3	4.4	4.4	(14)
(15)	Precipitation	PrecAvg	539	39	31	31	34	35	65	65	60	51	48	42	41	(15)
(16)		PrecMax	797	87	81	87	94	97	223	196	143	114	138	108	91	(16)
(17)		PrecMin	370	5	0	3	4	2	8	3	0	4	0	5	12	(17)
(18)		PrecStd	101	18	18	20	19	23	44	33	35	30	24	21	19	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.0	2.2	6.7	23.1	29.8	32.3	33.8	34.2	27.0	18.5	9.9	3.9	(19)
(20)			MCWB	1.2	0.9	3.3	13.3	17.6	20.0	19.4	20.0	16.8	12.0	8.9	3.4	(20)
(21)		2%	DB	1.0	1.2	4.2	20.1	27.0	29.9	31.1	30.9	24.0	15.1	7.4	2.1	(21)
(22)			MCWB	0.5	0.5	1.9	11.2	16.6	19.5	20.1	19.5	15.5	10.6	6.3	1.4	(22)
(23)		5%	DB	0.1	0.4	2.9	17.1	24.9	27.5	29.1	28.0	21.7	13.0	5.4	1.1	(23)
(24)			MCWB	-0.4	-0.1	1.1	10.4	15.4	18.7	19.7	18.8	14.8	9.7	4.7	0.5	(24)
(25)		10%	DB	-1.1	-0.9	1.8	14.0	22.2	25.2	27.2	25.7	19.1	11.1	3.8	0.1	(25)
(26)			MCWB	-1.5	-1.4	0.6	8.8	14.0	17.8	19.1	17.9	13.9	8.7	3.1	-0.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.2	1.2	3.7	14.1	18.9	22.2	22.8	21.7	18.2	13.3	8.8	3.1	(27)
(28)			MCDWB	1.4	1.6	6.1	21.2	26.7	28.9	28.7	28.9	24.0	16.5	9.6	3.4	(28)
(29)		2%	WB	0.4	0.5	2.1	12.4	17.5	20.7	21.7	20.6	16.7	11.3	6.5	1.4	(29)
(30)			MCDWB	0.7	0.9	3.6	18.6	25.2	27.3	27.9	27.7	21.9	14.2	7.1	1.8	(30)
(31)		5%	WB	-0.5	-0.1	1.3	10.7	16.2	19.7	20.8	19.6	15.4	10.1	4.6	0.5	(31)
(32)			MCDWB	-0.1	0.4	2.4	16.4	22.8	25.8	26.7	26.3	20.1	12.2	5.2	0.9	(32)
(33)		10%	WB	-1.8	-1.5	0.6	9.0	15.0	18.6	19.8	18.7	14.2	9.0	2.9	-0.3	(33)
(34)			MCDWB	-1.3	-1.0	1.5	14.0	20.8	24.1	25.3	24.4	18.5	11.0	3.4	0.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.1	6.9	7.2	8.7	10.3	9.9	9.8	8.3	5.9	4.5	5.5	(35)	
(36)			MCDBR	4.6	5.3	7.1	12.8	13.3	12.6	12.3	12.9	11.8	9.1	5.6	4.7	(36)
(37)			MCWBR	4.5	5.0	5.3	6.7	5.9	5.5	5.0	5.0	5.5	5.6	5.0	4.4	(37)
(38)		5% WB	MCDBR	4.7	5.1	6.3	12.2	12.2	11.2	10.8	11.2	10.5	8.0	5.1	4.5	(38)
(39)			MCWBR	4.7	4.9	5.0	6.9	6.0	5.6	5.1	4.9	5.6	5.5	4.8	4.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.290	0.318	0.362	0.408	0.388	0.380	0.401	0.428	0.375	0.349	0.312	0.286	(40)	
(41)		taud	2.061	2.049	2.010	2.120	2.321	2.387	2.334	2.234	2.362	2.327	2.188	2.043	(41)	
(42)		Ebn at Noon	583	719	778	805	853	864	835	780	777	696	564	483	(42)	
(43)		Edh at Noon	65	97	129	135	117	111	115	120	92	74	58	52	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.68	1.62	3.04	4.26	5.51	5.92	5.79	4.55	2.79	1.30	0.53	0.38	(44)	
(45)		RadStd	0.10	0.23	0.35	0.36	0.40	0.54	0.51	0.35	0.34	0.21	0.10	0.10	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.65	N/A	N/A	N/A	N/A	N/A	N/A	-220	+132	N/A
(47) Regional (8 neighbors)		+0.57	N/A	N/A	N/A	N/A	N/A	N/A	-168	+99	N/A

Nomenclature: See separate page